(1)	0	1	0
2	0	0	1
3.	0	0	0

$X_{n+1} = 1 \mid X_n = 1$

$\binom{4}{2}$ $\frac{4!}{\quad}$

(1) $\Omega = \{1, 2, 3, 4\}$.

$\mathcal{F} = \{\Omega, \phi, \{1\}, \{2,3,4\}, \{1,2\}, \{3,4\}\}$

(1) $\Omega = \{a, b, c, d\}$.

$\mathcal{F} = \{\Omega, \phi, \{a,b\}, \{b,c\}, \{c,d\}, \{ad\}, \{b,d\}, \{a,c\}\}$

$\{a,b\} \cup \{b,c\} = \{a,b,c\} \notin \mathcal{F}$

So, it is a λ system but not σ-field.

(2) $\Omega = \mathbb{R}$.

$\mathcal{F}$ = collection of all interval like $(-\infty, x)$

$\bigcup_{n=1}^{\infty} (-\infty, n) = (-\infty, \infty) = \Omega \in \mathcal{F}$ $\Rightarrow n = 0, 1, \ldots$

$\bigcap_{n=1}^{\infty} (-\infty, n) = (-\infty, 0) \in \mathcal{F}$

but, $(-\infty, 1)^c = [1, \infty] \notin \mathcal{F}$.

monotone class but not σ-field.

$\Omega = \{a, b, c, d\}$.

$\mathcal{F} = \{\phi, \{a\}, \{b, c\}, \Omega\}$

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(3) $\mathcal{F}$ be both π and λ system.

(a)

Consider,

$A_i \in \mathcal{F}$, A_i are not disjt.

define. $B_1 = A_1$

$B_2 = A_1 \cap A_2^c$

$\vdots$

$B_n = A_n \cap \left(\bigcup_1^{n-1} A_i\right)^c$

$= A_n \cap A_1^c \cap A_2^c \cdots A_{n-1}^c$.

by π-system B_i's $\in \mathcal{F}$.

B_i's are disjoint.

So, $\bigcup B_i \in \mathcal{F}$.

and $\bigcup B_i = \bigcup A_i \in \mathcal{F}$ (proved).

$\Omega, \emptyset \in \mathcal{F}$ by λ-system.

$A^c \in \mathcal{F}$ " "

(b) $\mathcal{F}$ be collection of both monotone class and field.

(1) $\Omega \in \mathcal{F}$ → field.

(2) $A^c \in \mathcal{F}$ → field.

(3) closed under countable union by monotone class.

$A_i \in \mathcal{F}$ → (1), (2) satisfied.

$A_1 \subseteq A_2 \subseteq \cdots A_n$

the $\bigcup_{n=1} A_i \in \mathcal{F}$.

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$\sigma(f^{-1}(A) : A \in \mathcal{F}) = \sigma(f^{-1}(A) : A \in \mathcal{B}(\mathcal{F}))$

$\sigma(f(\omega) : \omega \in \Omega) = f^{-1}(\mathcal{B}(\mathcal{F}))$

$= \theta \cdot e \ \underline{\theta \ln x} = \underline{\ln x}$

$= (f(A) : \omega \in \mathcal{B}(\mathcal{F}))$ $\mathcal{B}(\mathcal{F}) \subseteq \mathcal{F}$.

---

(2) $X : (\Omega, \mathcal{F}, P) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$

$\sigma(X) = \{ X^{-1}(A) : A \in \mathcal{B}_{\mathbb{R}} \}$

$\Rightarrow \quad \mathbb{R} \subseteq \mathcal{B}_{\mathbb{R}}$.

So, $X^{-1}(\mathbb{R}) \in \sigma(X)$

$\Rightarrow \Omega \in \sigma(X)$ —— (1)

if, $A \in \mathcal{B}_{\mathbb{R}}$.

$\Rightarrow A^c \in \mathcal{B}_{\mathbb{R}}$.

$X^{-1}(A^c) \in \sigma(X)$.

$\Rightarrow (X^{-1}(A))^c \in \sigma(X)$. —— (2)

$A_1, \ldots \in \mathcal{B}_{\mathbb{R}}$.

$\bigcup_{n=1}^{\infty} A_n \in \mathcal{B}_{\mathbb{R}}$.

$\Rightarrow \quad X^{-1}(\bigcup A_n) \in \sigma(X)$.

$\Rightarrow \quad \bigcup X^{-1}(A_n) \in \sigma(X)$ —— (3).

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(iv) $f^{-1}(\sigma(\mathcal{F})) = \{f^{-1}(A) : A \in \sigma(\mathcal{F})\}$

$\Omega' \in \sigma(\mathcal{F})$

$\Rightarrow f^{-1}(\Omega') \in f^{-1}(\sigma(\mathcal{F}))$.

$\Rightarrow \Omega \in f^{-1}(\sigma(\mathcal{F}))$ — (1)

$A^c \in \sigma(\mathcal{F})$.

$\Rightarrow f^{-1}(A^c) \in f^{-1}(\sigma(\mathcal{F}))$.

$\Rightarrow (f^{-1}(A))^c \in f^{-1}(\sigma(\mathcal{F}))$ — (2)

same for union.

$f^{-1}(\sigma(\mathcal{F}))$ is a σ-field.

$\mathcal{F} \subseteq \sigma(\mathcal{F})$

$\Rightarrow f^{-1}(\mathcal{F}) \subseteq f^{-1}(\sigma(\mathcal{F}))$.

good set principle.

$\Rightarrow \sigma(f^{-1}(\mathcal{F})) \subseteq f^{-1}(\sigma(\mathcal{F}))$ — (*)

$f^{-1}(\sigma(\mathcal{F})) \in f^{-1}(\mathcal{F}) \subseteq \sigma(f^{-1}(\mathcal{F}))$

(*)

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measure - 3

(8). $E \in \mathcal{B}_{\mathbb{R}}$. $A \subseteq E$ is open in E $\Leftrightarrow$ $B \subseteq \mathbb{R}$ open s.t $A = B \cap E$ show that $\mathcal{B}_E = \sigma(\text{all open sets in } E) = \mathcal{B}_{\mathbb{R}} \cap E$.

To show $\mathcal{B}_{\mathbb{R}} \cap E$ is a field $\longrightarrow$

(1) $E = \mathbb{R} \cap E$. $\in \mathcal{B}_{\mathbb{R}} \cap E$ $\longrightarrow$ $\mathbb{R} \cap E$

$E \in \mathcal{B}_{\mathbb{R}}$

(2) $A \in \mathcal{B}_{\mathbb{R}} \cap E$

A^c

(3) $A_1, \ldots \in \mathcal{B}_{\mathbb{R}} \cap E$.

$\bigcup_{n=1}^{\infty} A_n \in \mathcal{B}_{\mathbb{R}} \cap E$.

$\Rightarrow$ open sets in E $\subseteq \mathcal{B}_{\mathbb{R}} \cap E$, E

$\Rightarrow \sigma(\quad) \subseteq \mathcal{B}_{\mathbb{R}} \cap E$

$\Rightarrow \mathcal{B}_E \subseteq \mathcal{B}_{\mathbb{R}} \cap E$. (principle of good sets)

To show. $\mathcal{B}_{\mathbb{R}} \cap E \subseteq \mathcal{B}_E$,

//

4) (a) $\mathcal{A} := \{ A \in \sigma(\mathcal{F}) : P_1(A) = P_2(A) \}$

π system.

Let, if $\Omega \in \sigma(\mathcal{F})$ then

$P_1(\Omega) = 1$; $P_2(\Omega) = 1$

$P_1(\Omega) = P_2(\Omega)$.

$\Omega \in \mathcal{A}$. —— (1)

if $A \in \mathcal{A} \Rightarrow P_1(A) = P_2(A)$

$\Rightarrow 1 - P_1(A) = 1 - P_2(A)$

$\Rightarrow P_1(A^c) = P_2(A^c)$

$A^c \in \sigma(\mathcal{F})$.

So, $A^c \in \mathcal{A}$. —— (2)

• Suppose, $A_1, A_2, \ldots$ are disjoint $\in \mathcal{A}$. and $\in \sigma(\mathcal{F})$:

$\cup A_i \in \sigma(\mathcal{F})$.

$P_1(\cup A_i) = \sum P_1(A_i)$

$= \sum P_2(A_i)$

$= P_2(\cup A_i)$ —— (3).

$\mathcal{A}$ is a field.

$\mathcal{F} \subseteq \mathcal{A}$.

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